

Tarea 19

1.- Calcular los siguientes límites

$$a) \lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 4}{6x^2 + 2x - 1} = \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2} - \frac{3x}{x^2} + \frac{4}{x^2}}{\frac{6x^2}{x^2} + \frac{2x}{x^2} - \frac{1}{x^2}}$$

Se divide todo entre el mayor exponente (x^2)

$$\lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x} - \frac{4}{x^2}}{6 + \frac{2}{x} - \frac{1}{x^2}} = \frac{2 - \frac{3}{0} + \frac{4}{0}}{6 + \frac{2}{0} - \frac{1}{0}} = \frac{2}{6} = \frac{1}{3}$$

* Se sustituyen las x por ∞ porque no maneja, como vas a repartir un infinito XD

$$b) \lim_{x \rightarrow \infty} \frac{3x + 2}{x^3 + 4} = \lim_{x \rightarrow \infty} \frac{\frac{3x}{x^3} + \frac{2}{x^3}}{\frac{x^3}{x^3} + \frac{4}{x^3}}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{3}{x^2} + \frac{2}{x^3}}{1 + \frac{4}{x^3}} = \frac{0}{1} = 0$$

$$c) \lim_{x \rightarrow \infty} \frac{5h^4 - 2h^3 + 3}{3h^3 + 2h^4 + h} = \lim_{x \rightarrow \infty} \frac{\frac{5h^4}{h^4} - \frac{2h^3}{h^4} + \frac{3}{h^4}}{\frac{3h^3}{h^4} + \frac{2h^4}{h^4} + \frac{h}{h^4}}$$

$$\lim_{x \rightarrow \infty} \frac{5 - \frac{2}{h} + \frac{3}{h^4}}{\frac{3}{h} + \frac{2}{h^2} + \frac{1}{h^3}} = \frac{5}{0} \rightarrow \infty$$

2- Obtener la asíntota horizontal de las siguientes Funciones:

$$a) f(x) = \frac{3x+1}{2x+3} = \frac{\frac{3x^1}{x} + \frac{1}{x}}{\frac{2x^1}{x} + \frac{3}{x}} = \frac{3(1) + 0}{2(1) + 0} = \frac{3}{2}$$

$$b) y = \frac{x}{x^2+1} = \frac{\frac{x^1}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}} = \frac{0}{1+0} = \frac{0}{1} = 0$$

$$c) g(x) = \frac{x^2+1}{x-3} = \frac{\frac{x^2}{x^2} + \frac{1}{x^2}}{\frac{x}{x^2} - \frac{3}{x^2}} = \frac{1+0}{0-0} = \frac{1}{0}$$

$\therefore$ NO tiene asíntota